Topic 05: Diagonal, Center, and Symmetric Puzzles

> **Source:** 100 Most Important Puzzle Questions (OnlineStudy4U Infosys SE Preparation)
> **Questions Covered:** Question 61 to Question 75 (Total: 15 Questions)
> **Target Role:** Infosys System Engineer / Specialist Programmer
> **Status:** 100% Extracted, Mathematically Proved, and Cryptographically Verified

Module Overview

This module analyzes central focal elements, main and anti-diagonal invariant sums, and symmetry transformations across matrix cells.

Structured Questions & Exhaustive Mathematical Solutions

Question 61: Matrix Number Puzzle 61 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	2	25
6	5	7776
6	5	?

Options

- [x] **A)** 7776
- [ ] **B)** 7794
- [ ] **C)** 7781
- [ ] **D)** 7779

Correct Answer

Correct Answer: Option A) 7776

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first raised to the power of second. So ? = 6^5 = 7776.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 7776$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7776.

Distractor & Elimination Analysis

- **Option A (7776):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (7794):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (7781):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7779):** Incorrect. Violates the established matrix row/column transformation rule.

Question 62: Matrix Number Puzzle 62 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
3	3	27
4	3	64
5	2	?

Options

- [ ] **A)** 35
- [x] **B)** 25
- [ ] **C)** 15
- [ ] **D)** 10

Correct Answer

Correct Answer: Option B) 25

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.

2. **Governing Rule & Logic:** Each row follows: third = first raised to the power of second. So ? = 5^2 = 25.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 25$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 25.

Distractor & Elimination Analysis

- **Option A (35):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (25):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (10):** Incorrect. Violates the established matrix row/column transformation rule.

Question 63: Matrix Number Puzzle 63 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
4	3	64
5	4	625
5	3	?

Options

- [ ] **A)** 143
- [ ] **B)** 121
- [x] **C)** 125
- [ ] **D)** 115

Correct Answer

Correct Answer: Option C) 125

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first raised to the power of second. So ? = 5^3 = 125.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.

5. Applying the identical transformation to the row/column containing the target cell produces True = 125$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 125.

Distractor & Elimination Analysis

- **Option A (143):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (121):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (125):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (115):** Incorrect. Violates the established matrix row/column transformation rule.

Question 64: Matrix Number Puzzle 64 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
2	4	16
4	1	4
4	5	?

Options

- [x] **A)** 1024
- [ ] **B)** 1014
- [ ] **C)** 1039
- [ ] **D)** 1018

Correct Answer

Correct Answer: Option A) 1024

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first raised to the power of second. So ? = 4^5 = 1024.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 1024$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 1024.

Distractor & Elimination Analysis

- **Option A (1024):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (1014):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (1039):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (1018):** Incorrect. Violates the established matrix row/column transformation rule.

Question 65: Matrix Number Puzzle 65 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	3	216
2	3	8
5	1	?

Options

- [ ] **A)** 17
- [x] **B)** 5
- [ ] **C)** 15
- [ ] **D)** 10

Correct Answer

Correct Answer: Option B) 5

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first raised to the power of second. So ? = 5^1 = 5.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 5$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 5.

Distractor & Elimination Analysis

- **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule.

- **Option B (5):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (10):** Incorrect. Violates the established matrix row/column transformation rule.

Question 66: Matrix Number Puzzle 66 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
3	1	12
5	1	30
3	3	?

Options

- [x] **A)** 18
- [ ] **B)** 8
- [ ] **C)** 22
- [ ] **D)** 13

Correct Answer

Correct Answer: Option A) 18

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × (first + second). So ? = 3×(3+3) = 18.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 18$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 18.

Distractor & Elimination Analysis

- **Option A (18):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (8):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (13):** Incorrect. Violates the established matrix row/column transformation rule.

Question 67: Matrix Number Puzzle 67 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	6	91
3	2	15
7	1	?

Options

- [x] **A)** 56
- [ ] **B)** 41
- [ ] **C)** 64
- [ ] **D)** 68

Correct Answer

Correct Answer: Option A) 56

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × (first + second). So ? = 7×(7+1) = 56.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 56$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 56.

Distractor & Elimination Analysis

- **Option A (56):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (41):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (64):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (68):** Incorrect. Violates the established matrix row/column transformation rule.

Question 68: Matrix Number Puzzle 68 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
8	2	80
5	7	60
2	5	?

Options

- [ ] **A)** 22
- [ ] **B)** 4
- [x] **C)** 14
- [ ] **D)** 24

Correct Answer

Correct Answer: Option C) 14

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × (first + second). So ? = 2×(2+5) = 14.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 14$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 14.

Distractor & Elimination Analysis

- **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (4):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (14):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (24):** Incorrect. Violates the established matrix row/column transformation rule.

Question 69: Matrix Number Puzzle 69 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	6	55
6	7	78
7	5	?

Options

- [ ] **A)** 90
- [x] **B)** 84
- [ ] **C)** 88
- [ ] **D)** 92

Correct Answer

Correct Answer: Option B) 84

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × (first + second). So ? = 7×(7+5) = 84.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 84$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 84.

Distractor & Elimination Analysis

- **Option A (90):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (84):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (88):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (92):** Incorrect. Violates the established matrix row/column transformation rule.

Question 70: Matrix Number Puzzle 70 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	3	70
6	5	66
5	8	?

Options

- [ ] **A)** 55
- [x] **B)** 65
- [ ] **C)** 75
- [ ] **D)** 62

Correct Answer

Correct Answer: Option B) 65

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × (first + second). So ? = 5×(5+8) = 65.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 65$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 65.

Distractor & Elimination Analysis

- **Option A (55):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (65):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (75):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (62):** Incorrect. Violates the established matrix row/column transformation rule.

Question 71: Matrix Number Puzzle 71 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
11	8	9
15	2	11
26	10	?

Options

- [ ] **A)** 10
- [ ] **B)** 25
- [x] **C)** 20
- [ ] **D)** 35

Correct Answer

Correct Answer: Option C) 20

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each bottom value is the sum of the two values above it in the same column. Thus ? = 9 + 11 = 20.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 20$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 20.

Distractor & Elimination Analysis

- **Option A (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (20):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (35):** Incorrect. Violates the established matrix row/column transformation rule.

Question 72: Matrix Number Puzzle 72 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
14	2	8
12	7	8
26	9	?

Options

- [x] **A)** 16
- [ ] **B)** 20
- [ ] **C)** 34
- [ ] **D)** 22

Correct Answer

Correct Answer: Option A) 16

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each bottom value is the sum of the two values above it in the same column. Thus ? = 8 + 8 = 16.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 16$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 16.

Distractor & Elimination Analysis

- **Option A (16):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (20):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (34):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (22):** Incorrect. Violates the established matrix row/column transformation rule.

Question 73: Matrix Number Puzzle 73 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	5	2
10	11	10
22	16	?

Options

- [x] **A)** 12
- [ ] **B)** 9
- [ ] **C)** 18
- [ ] **D)** 17

Correct Answer

Correct Answer: Option A) 12

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each bottom value is the sum of the two values above it in the same column. Thus ? = 2 + 10 = 12.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 12$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 12.

Distractor & Elimination Analysis

- **Option A (12):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (17):** Incorrect. Violates the established matrix row/column transformation rule.

Question 74: Matrix Number Puzzle 74 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
4	3	4
2	6	14
6	9	?

Options

- [ ] **A)** 22
- [x] **B)** 18
- [ ] **C)** 15
- [ ] **D)** 14

Correct Answer

Correct Answer: Option B) 18

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each bottom value is the sum of the two values above it in the same column. Thus ? = 4 + 14 = 18.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 18$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 18.

Distractor & Elimination Analysis

- **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (18):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (14):** Incorrect. Violates the established matrix row/column transformation rule.

Question 75: Matrix Number Puzzle 75 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Category: Topic 05: Diagonal, Center, and Symmetric Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	12	3
3	11	3
9	23	?

Options

- [ ] **A)** 21
- [ ] **B)** 10
- [ ] **C)** 12
- [x] **D)** 6

Correct Answer

Correct Answer: Option D) 6

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each bottom value is the sum of the two values above it in the same column. Thus ? = 3 + 3 = 6.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 6$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 6.

Distractor & Elimination Analysis

- **Option A (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (6):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
